실전문제

1

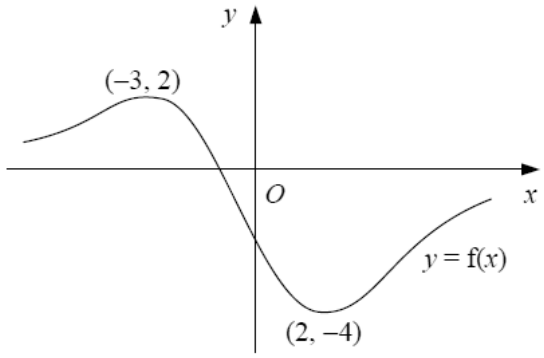


Figure 1

Figure 1 shows the curve $y = \mathrm{f}(x)$ which has a maximum point at $(-3, 2)$ and a minimum point at $(2, -4)$.

(a) Showing the coordinates of any stationary points, sketch on separate diagrams the graphs of

(i) $y = \mathrm{f}(|x|)$,

(ii) $y = 3\mathrm{f}(2x)$. **(7)**

(b) Write down the values of the constants a and b such that the curve with equation $y = a + \mathrm{f}(x + b)$ has a minimum point at the origin O. **(2)**

2 *(a)* Sketch the graph of $y = 2 + \sec\left(x - \frac{\pi}{6}\right)$ for x in the interval $0 \le x \le 2\pi$.

Show on your sketch the coordinates of any turning points and the equations of any asymptotes. **(5)**

(b) Find, in terms of π, the x-coordinates of the points where the graph crosses the x-axis. **(5)**

3 *(a)* Sketch on the same diagram the graphs of $y = 4a^2 - x^2$ and $y = |2x - a|$, where a is a positive constant. Show, in terms of a, the coordinates of any points where each graph meets the coordinate axes. **(6)**

(b) Find the exact solutions of the equation

$$4 - x^2 = |2x - 1|.$$ **(6)**

4

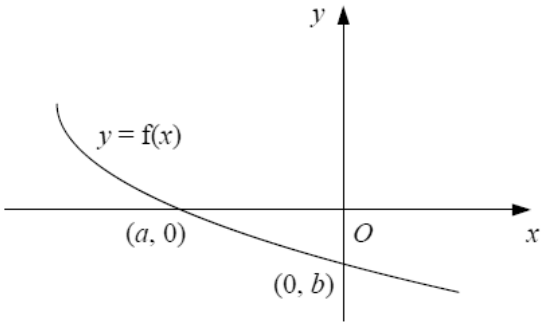


Figure 1

Figure 1 shows the graph of $y = \mathrm{f}(x)$ which meets the coordinate axes at the points $(a, 0)$ and $(0, b)$, where a and b are constants.

(a) Showing, in terms of a and b, the coordinates of any points of intersection with the axes, sketch on separate diagrams the graphs of

(i) $y = \mathrm{f}^{-1}(x)$,

(ii) $y = 2\mathrm{f}(3x)$. **(6)**

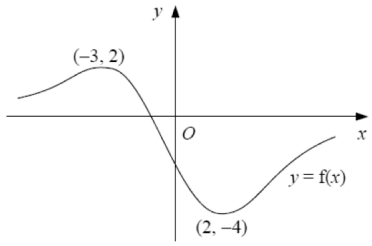
Given that

$$\mathrm{f}(x) = 2 - \sqrt{x+9}, \quad x \in \mathbb{R}, \quad x \geq -9,$$

(b) find the values of a and b, **(3)**

(c) find an expression for $\mathrm{f}^{-1}(x)$ and state its domain. **(5)**

1



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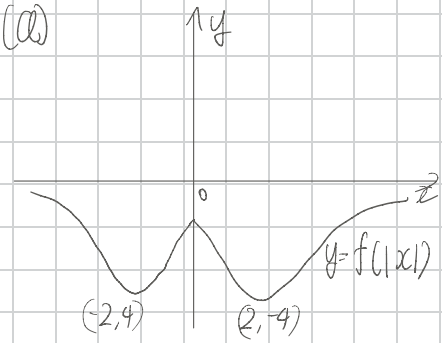

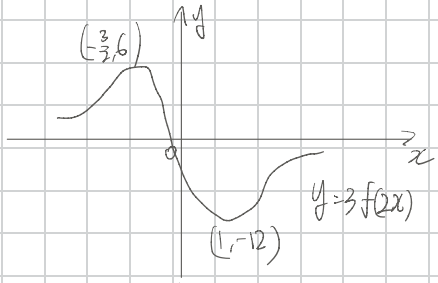

(b) minimum point of f(x) is (2, -4)

so, translate 2units to left, 4units to upward.

$y = 4 + f(x+2)$

$a = 4 \quad b = 2$

2 *(a)* Sketch the graph of $y = 2 + \sec\left(x - \frac{\pi}{6}\right)$ for x in the interval $0 \le x \le 2\pi$.

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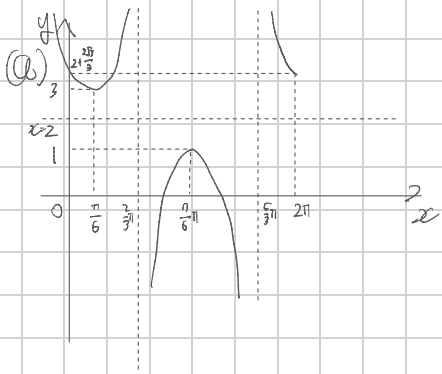
(b) Find, in terms of π, the x-coordinates of the points where the graph crosses the x-axis. **(5)**

(a)



(b) $2 + \sec\left(x - \frac{\pi}{6}\right) = 0$

$\sec\left(x - \frac{\pi}{6}\right) = -2$

$\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$

$= \frac{2}{3}\pi$ or $\frac{4}{3}\pi$

$\boxed{\frac{5}{6}\pi, \frac{3}{2}\pi}$

3 *(a)* Sketch on the same diagram the graphs of $y = 4a^2 - x^2$ and $y = |2x - a|$, where a is a positive constant. Show, in terms of a, the coordinates of any points where each graph meets the coordinate axes. **(6)**

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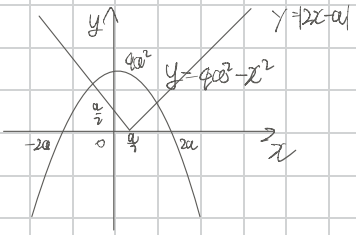
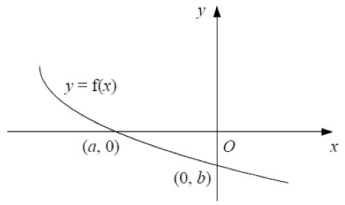


(b) $4-x^2=2x-1$, $x>\frac{1}{2}$, $x^2+2x-5=0$ $\frac{-2\pm\sqrt{4+20}}{2}=\frac{-2\pm2\sqrt{6}}{2}=\boxed{-1+\sqrt{6}}$

$4-x^2=-2x+1$, $x<\frac{1}{2}$, $x^2-2x-3=0$ $(x-3)(x+1)=0$ $\boxed{x=-1}$

4



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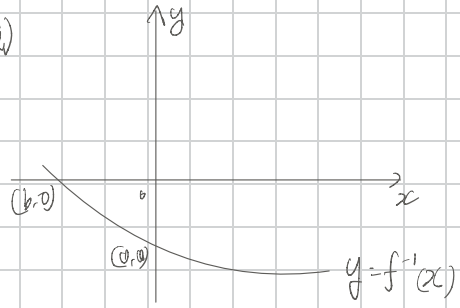
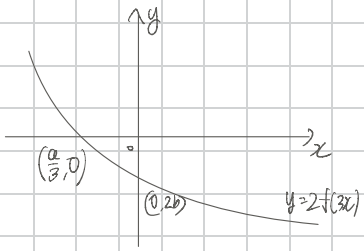
(a)





(b) $f(x)=2-\sqrt{x+9}$

$f(a)=0=2-\sqrt{a+9}$

$\sqrt{a+9}=2$

$a+9=4$ $\boxed{a=-5}$

$f(0)=b=2-\sqrt{9}$

$\boxed{b=-1}$

(c) $y=2-\sqrt{x+9}$

$2-y=\sqrt{x+9}$

$(2-y)^2-9=x$ $f^{-1}(x)=(x-2)^2-9$, $x\in R$